

APEX SPC-RPN AI RISK ASSESSMENT

SPC-RPN Risk Assessment

Data-Grounding Note (Required for Transparency)

The combined report provides mean, UCL, LCL, Sigma_Level, RPN, and a "signals" count per metric_id — it does not include raw time-series data, so specific control-chart rule violations (e.g., "Run of 8," "2-of-3 beyond 2σ") cannot be confirmed or named from this dataset; only the *aggregate signal count* is available as a proxy for instability. All 218 metrics report an identical Sigma_Level of 3.0, meaning control limits were uniformly set at $\pm 3\sigma$ — this is a limit-setting convention, not itself evidence of stability or instability.

A structural pattern was confirmed in the source data: $RPN = signals \times 35$ holds exactly across every row (e.g., PBI10000: $59 \times 35 = 2065$; BCB10000: $10 \times 35 = 350$). This means the "signals" field is the sole differentiator driving RPN ranking — Severity and Detection sub-scores are not separately reported in this dataset. Per grounding rules, S/O/D values below are marked "Not decomposed in source" rather than invented.

Statistical Process Control Findings

Out-of-Control / Highest-Signal Processes ($RPN \geq 1500$, signals ≥ 43):

Metric ID	Mean	UCL	LCL	Signals	RPN
PBI10000	50.15	70.06	30.23	59	2065
RCD10000	60.08	72.10	48.07	54	1890
ERS10000	42.49	52.70	32.27	53	1855
CPG10000	45.02	55.21	34.83	50	1750
FLO10000	64.01	77.84	50.18	50	1750
PPB10000	50.55	72.70	28.40	50	1750
BSP10000	50.06	72.47	27.65	46	1610
RST10000	60.10	72.07	48.13	45	1575
LOER_001	60.62	61.83	59.42	44	1540
TQA10000	56.54	69.22	43.86	44	1540
RSS10000	50.23	68.68	31.79	43	1505
SCP10000	30.00	36.55	23.44	43	1505
SRVC_001	29.99	31.69	28.29	43	1505

These 13 modules carry the highest signal density in the enterprise dataset and should be treated as the priority instability cluster. Note: LOER_001 and SRVC_001 show unusually narrow UCL/LCL bands relative to mean (e.g., SRVC_001: LCL 28.29–UCL 31.69, a spread of only ~3.4), suggesting tighter process tolerance — even moderate variation here could generate disproportionate signal counts.

Processes Trending Toward Instability (Watch Tier, RPN 1100–1470 / signals 32–42):

This is a large band (~60 modules) transitioning from moderate to high risk, including ANO10000, CUST_001, MSP10000, RSL10000, CII10000, ROR10000, CLC10000, CTP10000, PUR10000, CSP10000, EFI10000, EHU10000, ESC10000, RMR10000, BPA10000, CCM10000, CNT10000, EAU10000, EGV10000, SCC10000, and others in the 1100–1470 RPN range. Direction: these modules sit just below the critical-signal threshold (43 signals); a further increase of even 1–2 signal events per cycle would push several (e.g., CII10000 at 41 signals, ROR10000 at 41 signals) into the Out-of-Control tier. Estimated time-to-violation cannot be quantified precisely without trend history across multiple reporting periods — only a single snapshot date (2026-09-27) is provided.

Comparatively Stable Processes (RPN < 700, signals < 20):

GRC10000 (665/19), DAM10000 & PFR10000 (595/17), BGL10000 (525/15), BAL10000/BLC10000/CGS10000 (490/14), IGV10000 (455/13), BTP10000 (385/11), BCB10000 (350/10) — lowest observed signal counts, indicating comparatively better control performance within this dataset.

Risk Priority Number Analysis

| Failure Mode | Module | Severity | Occurrence | Detection | RPN | Action Required |

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| Recurrent out-of-control signal generation (highest frequency in dataset) | PBI10000 | Not decomposed in source | 59 signals (highest) | Not decomposed in source | 2065 | Immediate root-cause investigation; escalate to process owner |

| High-frequency control-limit signal activity | RCD10000 | Not decomposed in source | 54 signals | Not decomposed in source | 1890 | Priority review within current cycle |

| High-frequency control-limit signal activity | ERS10000 | Not decomposed in source | 53 signals | Not decomposed in source | 1855 | Priority review within current cycle |

| Elevated signal count, tied cluster | CPG10000 | Not decomposed in source | 50 signals | Not decomposed in source | 1750 | Schedule root-cause analysis |

| Elevated signal count, tied cluster | FLO10000 | Not decomposed in source | 50 signals | Not decomposed in source | 1750 | Schedule root-cause analysis |

| Elevated signal count, tied cluster | PPB10000 | Not decomposed in source | 50 signals | Not decomposed in source | 1750 | Schedule root-cause analysis |

| Narrow-tolerance instability (tight UCL/LCL band) | SRVC_001 | Not decomposed in source | 43 signals | Not decomposed in source | 1505 | Review specification limits vs. natural process variation |

| Narrow-tolerance instability (tight UCL/LCL band) | LOER_001 | Not decomposed in source | 44 signals | Not decomposed in source | 1540 | Review specification limits vs. natural process variation |

Note: Severity and Detection ratings require inputs (impact assessment, current detection method effectiveness) not present in the source report and are therefore not fabricated here. Only Occurrence (signals) and total RPN are grounded in the data.

Critical Corrective Actions (Ranked by RPN)

1. Investigate PBI10000 Signal Recurrence - RPN: 2065 - Due: Immediate (within 5 business days)

- Root Cause (HYPOTHESIS): Process mean (50.15) sits near the midpoint of a wide control band (30.23–70.06); the high signal count (59) suggests either elevated variance in the underlying data-generating process or a measurement/detection artifact inflating signal detection frequency — requires investigation, not confirmed.
- Corrective Action: Pull underlying time-series data for PBI10000; run Western Electric rule analysis to classify actual violation types (trend, shift, run) rather than relying on aggregate signal count.
- Verification Method: Re-run SPC control chart after root-cause fix; confirm signal count reduction in next reporting cycle.

2. Investigate RCD10000 and ERS10000 (Tied High-Signal Cluster) - RPN: 1890 / 1855 - Due: 10 business days

- Root Cause (HYPOTHESIS): Both modules show signal counts in the low-50s, second only to PBI10000; possible shared upstream data source or common process input driving correlated instability — hypothesis only, not verified.
- Corrective Action: Cross-reference data pipelines/owners for RCD10000 and ERS10000 to test shared-cause hypothesis; if confirmed, remediate at source.
- Verification Method: Compare signal counts pre/post remediation across both modules simultaneously to test for correlated improvement.

3. Address Tied-RPN Cluster (CPG10000, FLO10000, PPB10000) - RPN: 1750 each - Due: 15 business days

- Root Cause (HYPOTHESIS): Identical RPN/signal counts (50 signals each) across three distinct modules may indicate a shared reporting/measurement system rather than three independent process failures — requires verification.
- Corrective Action: Audit whether these three modules share a common data collection or calculation methodology before treating as independent failure modes.
- Verification Method: Trace signal-generation logic per module; confirm independence or shared root cause.

4. Review Narrow-Tolerance Modules (SRVC_001, L0ER_001) - RPN: 1505 / 1540 - Due: 15 business days

- Root Cause (HYPOTHESIS): Both modules exhibit unusually tight UCL–LCL spreads relative to peer modules, which mechanically increases the probability of signal generation from normal variation — hypothesis, requires confirmation against original specification rationale.
- Corrective Action: Re-validate whether current control limits reflect true process capability or overly conservative specification limits; recalculate control limits if warranted.
- Verification Method: Compare recalculated Cpk/Ppk against signal frequency post-adjustment.

5. Monitor Watch-Tier Modules (RPN 1100–1470) - RPN: up to 1470 - Due: Next quarterly review cycle

- Root Cause (HYPOTHESIS): Modules approaching the 43-signal critical threshold may reflect gradual degradation not yet flagged as critical — trend requires multi-period data to confirm, which is not available in this single snapshot.
- Corrective Action: Flag for trend tracking in next reporting cycle; request historical SPC data to establish trajectory.
- Verification Method: Compare signal counts across consecutive reporting periods once available.

Process Stability Summary

Across the 218 modules assessed, all report a uniform 3.0 Sigma control-limit setting, with RPN values ranging from 350 (BCB10000) to 2065 (PBI10000) — a span driven entirely by differences in detected signal counts (10 to 59), since $RPN = \text{signals} \times 35$ holds consistently across the dataset. Thirteen modules ($RPN \geq 1500$) represent a concentrated high-risk cluster requiring immediate root-cause investigation, while a broad watch-tier band (~60 modules, RPN 1100–1470) warrants trend monitoring to prevent escalation. Because the source report lacks raw time-series data and explicit control-chart rule labels, all root causes above are stated as hypotheses requiring further investigation, and Severity/Detection RPN components could not be decomposed from the available data.